

Homework #5

[1] 已知 $\sqrt{n+a} - \sqrt{n} \approx \frac{a}{2\sqrt{n}}$

证明: $f(x) = \sqrt{n+a}$, $g(x) = \sqrt{n}$

$$df = f'(x) dx$$

设 $n+a = x \Rightarrow f'(x) = \frac{1}{2\sqrt{x}}$

small $a \leftrightarrow dx = a$

$\hookrightarrow n \rightarrow \infty, x \rightarrow \infty \leftrightarrow n=x$

$$df = \frac{a}{2\sqrt{n}}$$

$$\Rightarrow \sqrt{n+a} - \sqrt{n} \approx df \approx \frac{a}{2\sqrt{n}}$$

$$\sqrt{n+a} - \sqrt{n} \approx \frac{a}{2\sqrt{n}} \text{ 成立}$$

计算 $n = 10^{26}, a = 5$

$$\sqrt{10^{26}+5} - \sqrt{10^{26}} \approx \frac{5}{2 \times 10^{13}} = 2.5 \times 10^{-13}$$

[2]

1) $z = (x+y)^5$

$$\frac{dz}{dx} = \frac{\partial z}{\partial y} \cdot \frac{dy}{dx} + \frac{\partial z}{\partial x}$$

$$\Rightarrow \frac{\partial z}{\partial y} = 5(x+y)^4$$

$$y = \sin(10x)$$

$$\frac{dy}{dx} = 10 \cos(10x)$$

$$\begin{aligned} \Rightarrow \frac{dz}{dx} &= 5(x+y)^4 \cdot 10 \cos(10x) + 5(x+y)^4 \\ &= 5(x+y)^4 (1 + 10 \cos(10x)) \end{aligned}$$

$$2) c = \sin(a-b), \quad b = ae^{2a}$$

$$\frac{dc}{da} = \frac{\partial c}{\partial b} \frac{db}{da} + \frac{\partial c}{\partial a}$$

$$\Rightarrow \frac{\partial c}{\partial b} = -\cos(a-b)$$

$$\frac{db}{da} = 2ae^{2a} + e^{2a} = 2b + e^{2a}$$

$$\frac{\partial c}{\partial a} = \cos(a-b)$$

$$\Rightarrow \frac{dc}{da} = (1 - 2b - e^{2a}) \cos(a-b)$$

[3]

$$1) z = x \cdot e^{-y}, x = \cosh t, y = \cos s$$

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s}$$

$$= -x e^{-y} (-\sin s)$$

$$= z \cdot \sin s$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}$$

$$= e^{-y} \cdot \sinh t$$

$$2) w = e^{-r^2 - s^2}, r = uv, s = u + 2v$$

$$\frac{\partial w}{\partial u} = \frac{\partial w}{\partial r} \frac{\partial r}{\partial u} + \frac{\partial w}{\partial s} \frac{\partial s}{\partial u}$$

$$= -2r e^{-r^2 - s^2} v + (-2s) e^{-r^2 - s^2} \cdot 1$$

$$= -2(rv + s) w$$

$$\frac{\partial w}{\partial v} = \frac{\partial w}{\partial r} \frac{\partial r}{\partial v} + \frac{\partial w}{\partial s} \frac{\partial s}{\partial v}$$

$$= -2r e^{-r^2 - s^2} \cdot u + (-2s) e^{-r^2 - s^2} \cdot 2$$

$$= -2(ru + 2s) \cdot w$$

$$3) w = x + y, x^2 + xy + y^2 = s, x^2 y + xy^2 = t$$

$$\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x} \frac{dx}{ds} + \frac{\partial w}{\partial y} \frac{dy}{ds}$$

$$\Rightarrow 3x^2 dx + y dx + x dy + 3y^2 dy = ds$$

$$2xy dx + x^2 dy + y^2 dx + 2yx dx = dt$$

$$\Rightarrow (3x^2 + y) dx + (x + 3y^2) dy = ds$$

$$(2xy + y^2) dx + (2xy + x^2) dy = dt$$

$$dz = dx + dy$$

$$dx = \frac{\begin{vmatrix} ds & x+3y^2 \\ dt & 2xy+x^2 \end{vmatrix}}{\begin{vmatrix} 3x^2+y & x+3y^2 \\ 2xy+y^2 & 2xy+x^2 \end{vmatrix}}, \quad dy = \frac{\begin{vmatrix} 3x^2+y & ds \\ 2xy+y^2 & dt \end{vmatrix}}{\begin{vmatrix} \sim & \sim \end{vmatrix}}$$

$$\Rightarrow dz = ds \frac{x^2 - y^2}{|\sim|} + dt \frac{3x^2 y - x - 3y^2}{|\sim|}$$

$$|\sim| = 6x^3 y + 2xy^2 + 3x^4 + x^2 y - (2x^2 y + xy^2 + 6xy^3 + 3y^4)$$

$$= 3x^4 + 6x^3 y + xy^2 - x^2 y - 6xy^3 - 3y^4$$

$$= (x-y) 3(x+y)^3 - xy(x-y)$$

$$= (x-y) (3w^3 - xy)$$

$$\frac{\partial z}{\partial s} = \frac{(x-y)(x+y)}{(x-y)(3w^2-xy)} = \frac{w}{3w^2-xy}$$

$$\frac{\partial z}{\partial t} = \frac{(3(x+y)-1)(x-y)}{(x-y)(3w^2-xy)} = \frac{3w-1}{3w^2-xy}$$

$$[4] \quad 2x+3y+z-1=0, \quad 4x^2+y^2+z^2=0$$

$$f(x, y, z) = 4x^2 + y^2 + z^2 = 0$$

$$\varphi(x, y, z) = 2x + 3y + z - 1 = 0$$

$$\Rightarrow df = 8x dx + 2y dy + 2z dz = 0$$

$$d\varphi = 2dx + 3dy + dz = 0$$

$$\Rightarrow \frac{\partial F}{\partial x} = 8x + 2\lambda = 0$$

$$\frac{\partial F}{\partial y} = 2y + 3\lambda = 0$$

$$\frac{\partial F}{\partial z} = 2z + \lambda = 0$$

$$\Rightarrow 2\left(-\frac{\lambda}{4}\right) + 3\left(-\frac{2}{3}\lambda\right) + \left(-\frac{\lambda}{2}\right) = 1$$

$$-1 - \frac{9}{2} \lambda = 1$$

$$\lambda = -2$$

$$\Rightarrow x = \frac{1}{2}, y = 3, z = 1$$

$$\Rightarrow (\frac{1}{2}, 3, 1)$$

$$[5] \quad 2x + y - z = 1, x - y + z = 2$$

$$F = x^2 + y^2 + z^2 + \lambda_1 (2x + y - z) + \lambda_2 (x - y + z) -$$

$$\begin{cases} 2x + 2\lambda_1 + \lambda_2 = 0 \\ 2y + \lambda_1 - \lambda_2 = 0 \\ 2z - \lambda_1 + \lambda_2 = 0 \end{cases}$$

$$x = \frac{-2\lambda_1 - \lambda_2}{2}, y = \frac{-\lambda_1 + \lambda_2}{2}, z = \frac{\lambda_1 - \lambda_2}{2}$$

$$\begin{cases} -2\lambda_1 - \lambda_2 + \lambda_2 - \lambda_1 = 1 \end{cases}$$

$$\begin{cases} -2\lambda_1 - \lambda_2 + \lambda_1 - \lambda_2 + \lambda_1 - \lambda_2 = 4 \end{cases}$$

$$\Rightarrow \begin{cases} \lambda_1 = -\frac{1}{3} \\ \lambda_2 = -\frac{4}{3} \end{cases} \Rightarrow x = 1, y = -\frac{1}{2}, z = \frac{1}{2}$$

$$d = \sqrt{1 + \frac{1}{4} + \frac{1}{4}} = \frac{\sqrt{6}}{2}$$

$$[6] \quad \frac{\partial^2 z}{\partial x^2} - 5 \frac{\partial^2 z}{\partial x \partial y} + 6 \frac{\partial^2 z}{\partial y^2} = 0$$

$$s = y + 2x, t = y + 3x \Rightarrow \frac{\partial^2 z}{\partial s \partial t} = 0$$

$$\Rightarrow \frac{\partial s}{\partial x} = 2, \frac{\partial s}{\partial y} = 1$$

$$\frac{\partial t}{\partial x} = 3, \frac{\partial t}{\partial y} = 1$$

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial s} \cdot \frac{\partial s}{\partial x} + \frac{\partial z}{\partial t} \frac{\partial t}{\partial x}, \quad \frac{\partial z}{\partial y} = \frac{\partial z}{\partial s} + \frac{\partial z}{\partial t}$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left(2 \frac{\partial z}{\partial s} + 3 \frac{\partial z}{\partial t} \right) \quad \frac{\partial z}{\partial x} \left(2 \frac{\partial^2 z}{\partial s^2} + 3 \frac{\partial^2 z}{\partial s \partial t} \right) - 5 \frac{\partial z}{\partial x} \left(\frac{\partial^2 z}{\partial s^2} + \frac{\partial^2 z}{\partial s \partial t} \right)$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial s} + \frac{\partial z}{\partial t} \right) \Rightarrow \quad + 6 \frac{\partial z}{\partial y} \left(\frac{\partial^2 z}{\partial s^2} + \frac{\partial^2 z}{\partial s \partial t} \right) = 0$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial s} + \frac{\partial z}{\partial t} \right) \quad \hookrightarrow - \frac{\partial^2 z}{\partial s \partial t} = 0$$

$$\hookrightarrow \frac{\partial^2 z}{\partial s \partial t} = 0$$

$\Rightarrow$ 代入原方程

$$\frac{\partial^2 z}{\partial s \partial t} = 0 \Rightarrow z = f(t)s + g(t)$$

$$f(t) = \frac{\partial z}{\partial t}$$

$$[7] \quad \frac{\partial x}{\partial u} = 1, \quad \frac{\partial x}{\partial v} = 1$$

$$\frac{\partial y}{\partial u} = 1, \quad \frac{\partial y}{\partial v} = -1$$

$$\frac{\partial w}{\partial u} = \frac{\partial w}{\partial x} + \frac{\partial w}{\partial y}$$

$$\frac{\partial w}{\partial v} = \frac{\partial w}{\partial x} - \frac{\partial w}{\partial y}$$

$$\frac{\partial^2 w}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial w}{\partial x} \right), \quad \frac{\partial^2 w}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial w}{\partial y} \right)$$

$$\frac{\partial w}{\partial u} \left(\frac{\partial w}{\partial u} + \frac{\partial w}{\partial v} \right) \cdot \frac{1}{\partial x} + \frac{\partial w}{\partial v} \left(\frac{\partial w}{\partial u} + \frac{\partial w}{\partial v} \right) \cdot \frac{1}{\partial x}$$

$$= \frac{\partial^2 w}{\partial u^2} + \frac{\partial^2 w}{\partial v^2} + \frac{\partial^2 w}{\partial u \partial v} - \frac{\partial^2 w}{\partial u^2} - \frac{\partial^2 w}{\partial v^2} = 1$$

$$\Rightarrow \frac{\partial^2 w}{\partial u \partial v} = 1 \Rightarrow w = \frac{1}{2} u^2 + u g(v) + h(u)$$

[8]

$$c1) y^{\ln x} = y \ln^y$$

$$\frac{dy}{y \ln^y} = \frac{dx}{\sin x}$$

$$\ln \ln^y + C = \ln |\tan \frac{x}{2}|$$

$$y = e, \quad x = \frac{\pi}{3}$$

$$C = \ln \frac{\sqrt{3}}{3}$$

$$12) y dy + (xy^2 - 8x) dx = 0$$

$$y=3 \Rightarrow y^2=8 \Rightarrow \frac{y}{y^2-8} dy = -x dx$$

$$\frac{\ln y^2-8}{2} + C = -\frac{1}{2}x^2$$

$$\Rightarrow \text{constant } C = -\frac{1}{2}$$

对于 $8-y^2$ 的

也解 $y=3$

某情况舍去

$$\Rightarrow \text{constant } C = -\frac{1}{2}$$

[9]

$$c1) y'' - y' - 2y = 3e^x$$

$$r^2 - r - 2 = 0$$

$$(r-2)(r+1) = 0$$

$$y^x = Ae^{2x} + Be^{-x}$$

$$b=1 \Rightarrow \bar{y} = x Q(x) e^{2x}$$

$$Q(x) = a$$

$$\bar{y}' = a e^{2x} + 2ax e^{2x}$$

$$\bar{y}'' = 2a e^{2x} + (2a + 4ax) e^{2x}$$

$$3 = (4a + 4ax - a - 2ax - 2ax)$$

$$a = 1$$

$$\Rightarrow y = A e^{2x} + B e^{-x} + x e^{2x}$$

$$(2) 5y'' + 12y' + 20y = 120 \sin(2x)$$

$$5r^2 + 12r + 20 = 0$$

$$\Delta = -256 = (16i)^2$$

$$r = -1.2 \pm 1.6i$$

$$y^{\text{hom}} = e^{-1.2y} (A \cos 1.6y + i C_2 \sin 1.6y)$$

$$\Rightarrow \bar{y} = a \cdot e^{i2x}, \bar{y}' \sim, \bar{y}'' = \sim$$

$$\Rightarrow a = -5i$$

$$\Rightarrow y = e^{-\frac{6}{5}x} \left(A \sin\left(\frac{8}{5}x\right) + B \cos\left(\frac{8}{5}x\right) \right) - 5 \cos 2x$$

$$(3) (D^2 + 2D + 1)y = 60e^{-4x} \sin 5x$$

$$r^2 + 2r + 1 = 0 \quad \Delta = (-2)^2$$

$$r = -1 \pm 4i, \quad k=0$$

$$y^x = e^{-x} (A \sin 4x + B \cos 4x)$$

$$\bar{y} = a \cdot e^{(-4+5i)x}$$

$$\Rightarrow a = 2$$

$$\Rightarrow y = e^{-x} (A \sin 4x + B \cos 4x) + 2e^{-4x} \cos 5x$$

$$(4) 5y'' + 6y' + 2y = x^2 + 6x$$

$$5r^2 + 6r + 2 = 0, \quad \Delta = -4$$

$$r = -\frac{3}{5} \pm \frac{1}{5}i$$

$$y^x = e^{-\frac{3}{5}x} (A \sin \frac{x}{5} + B \cos \frac{x}{5})$$

$$\bar{y} = ax^2 + bx + c$$

$$\Rightarrow a = \frac{1}{5}, \quad b=0, \quad c=0$$

$$\Rightarrow y = e^{-\frac{3}{5}x} (A \sin(\frac{x}{5}) + B \cos(\frac{x}{5})) + \frac{x^2 + 6x}{5}$$

[10]

$$(D-1)^2 y = 4e^x + (1-x)(e^{2x}-1)$$

$$(r-1)^2 = 0 \Rightarrow r_1 = r_2 = 1$$

$$y^x = Ae^x + Bxe^x$$

$$\textcircled{1} y'' - 2y' + y = 4e^x$$

$$\bar{y} = ax^2 e^x \Rightarrow a = 2$$

$$\bar{y}(1) = 2x^2 e^x$$

$$\textcircled{2} y'' - 2y' + y = (1-x)e^{2x}$$

$$\bar{y} = (ax+b)e^{2x} \Rightarrow a = -1, b = 3$$

$$\bar{y}(1) = (3-x)e^{2x}$$

$$\textcircled{3} \bar{y} = ax+b \Rightarrow a = 1, b = 1$$

$$\bar{y}(1) = x+1$$

$$\Rightarrow y = (A+Bx)e^x + 2x^2 e^x + (3-x)e^{2x} + x+1$$